

Tutorial 08

Index No. 25000799

(a)

(a) A sequential circuit is a digital circuit whose outputs depends not only on the inputs, but also on the past sequence of inputs.

(b)

Combinational	Sequential
• output depends on current inputs. • No memory element • No feedback path	• output depends on current inputs as well as past state. • Has a memory element • Has a feedback path

(c) registers, counters, RAM

(d)

Synchronous	Asynchronous
• state changes controlled by a common clock signal • Easier to design • Uses flip-flops	• state changes happen as soon as inputs change. • Complex to design • uses latches.



(2)

(a) A clock is a periodic square wave signal used to synchronize the operation of sequential circuits.

(b) Latches - SR latch, D Latch

Flip flops - SR Flip Flop, D Flip Flop, JK Flip Flop, T Flip Flop

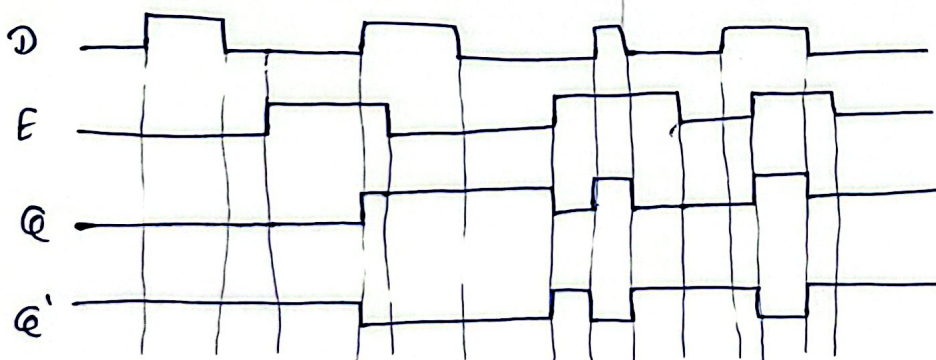
(c)

Flip Flops	Latches
• Edge triggered (changes only at clock edge) • Output changes only at clock transition • Safe for synchronous systems	• Level-triggered (transparent while enable is active) • Output can change any time. • more prone to timing hazards.

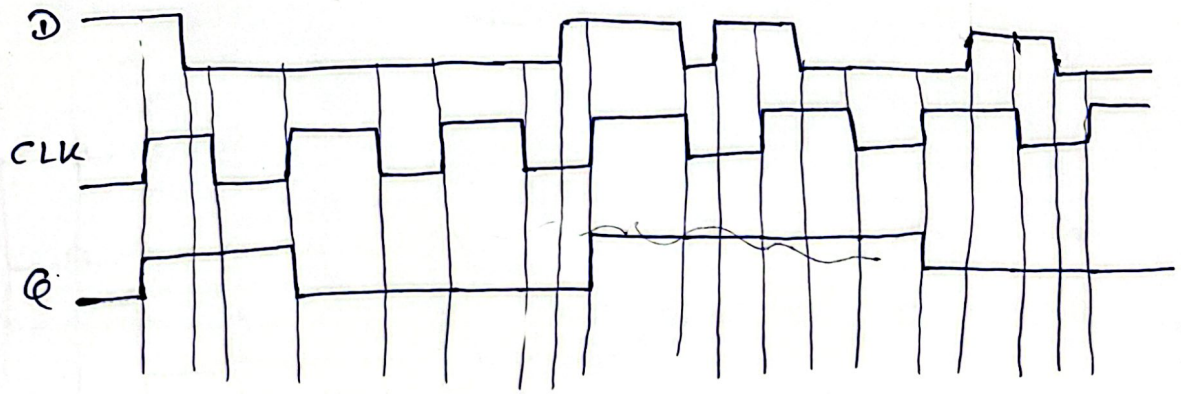
(d)

clock signal	Enable signal
• A clock signal is typically periodic and triggers state changes on its edges.	• An enable signal is usually level based. It doesn't need to be periodic

(3)



(4)



(5)

